MIL-V-36590A (DSA-DM)
19 May 1975
SUPERSEDING
MIL-V-0036590 (DSA-DM)
16 September 1969

MILITARY SPECIFICATION

VIEW BOX, RH BLOOD TYPING, 115 volt, AC-DC

This limited coordination Military specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military specification MIL-V-0036590 (DSA-DM). It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers a view box suitable for providing an optimum surface temperature (i.e. within a range of 45° to 50° Centigrade (°C.)) for use in the slide method Rh blood typing procedure (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping, Fiberboard.

Military

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center

Drawing No. 10173	View Box, Rh Blood Typing, 115 volt, AC-DC

(Copies of specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C73.11	Dimensions of Caps, Plugs and Receptacles, 125 volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type, General Purpose.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Frame and housing. The frame and housing shall be made of a durable, smoothly finished hardwood.

3.1.2 Metal components. All metal components, including screws, washers, nuts, nails, etc., shall be resistant to corrosion.

3.2 Design and construction. The view box shall conform to DPSC Drawing No. 10173 for design, construction and dimensions. The view box shall consist of the following:

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a. Stand
b. Housing
c. Observation glass
d. Lamp assembly
d. Lamp cord
e. "Off-on" switch

3.2.1 Stand. The stand shall consist of two uprights and a base. The base shall be mounted on four rubber feet. As indicated in DPSC Drawing No. 10173, the stand shall pivotally support the housing (see 3.2.2). The uprights shall be properly positioned so that the housing can tilt freely without any tendency for the housing to rub or bind against the uprights.

3.2.2 Housing. The housing shall enclose the lamp assembly (see 3.2.4) and shall accommodate the lamp cord (see 3.2.5) and "off-on" switch (see 3.2.6). The upper portion of the inside of the housing shall have mounting grooves (see 3.2.2.1) to support the observation glass (see 3.2.3) and microscope slides. The housing shall be supplied with a retaining strip (see 3.2.2.2) to secure the observation glass in the mounting grooves. When the observation glass is properly positioned in the mounting grooves, the housing shall be able to accommodate up to eight microscope slides (see 6.2) having the following maximum dimensions (millimeters):

Length----------76.6 mm.
Width----------25.8 mm.
Thickness----------1.35 mm.

Such microscope slides shall be able to be placed on the observation glass with their lengths normal to the length of the housing, and their ends engaged in the mounting grooves.

3.2.2.1 Glass mounting grooves. The mounting grooves in the housing shall be properly formed and oriented so that inserted slides can be positioned over the observation glass without interference. The width of the grooves (i.e. vertical dimensions)shall be able to accommodate all of the following:

a. The observation glass
b. Microscope slides having maximum thicknesses of 1.35 mm.
c. The retaining strip
d. The interchanging of observation glasses from other view boxes of the same design offered under the same contract or purchase order.

3.2.2.2 Retaining strip. A rectangular wood strip conforming to DPSC Drawing No. 10173 shall be supplied. The wood strip shall have sufficient length and thickness (i.e. height of strip within the mounting groove) to accomplish the following, when inserted in the mounting grooves at one end of the housing:

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a. The retaining strip can be press-fitted in the mounting grooves to retain the observation glass.

b. The retaining strip can be removed from the mounting grooves in order to replace components in the housing (e.g. observation glass or lamp bulb).

c. Microscope slides, having maximum thickness of 1.35 mm., can be inserted in the mounting grooves and over the observation glass without removing the retaining strip.

3.2.3 Observation glass. A blue opalite glass panel, conforming to DPSC Drawing No. 10173, shall be supplied. The observation glass shall be sufficiently flat so that, when it is in place in the housing, microscope slides can be positioned over the observation glass without interference. In addition, when positioned in the mounting grooves of the housing, the width of the observation glass shall be such that:

a. A slide fit exists within the mounting grooves.

b. Both sides of the glass panel engage the grooves of the housing without any tendency to slip out of position.

c. Interchangeability exists with glass panels from other view boxes of the same design that are offered under the same contract or purchase order.

3.2.4 Lamp assembly. The lamp assembly shall provide the light and heat source for the view box. The lamp assembly shall consist essentially of the following:

a. A tubular type frosted Lumiline lamp for operation from 115 volt AC-DC supply lines (see 6.3).

b. Lampholder components (base, caps, etc.) for securing the Lumiline lamp within the housing as indicated in DPSC Drawing No. 10173.

c. Electrical wiring and components for attaching the lamp cord (see 3.2.5) and "off-on" switch (see 3.2.6).

3.2.5 Lamp cord. A flexible 3-wire line cord with non-detachable plug, conforming to DPSC Drawing No. 10173, shall be supplied. The plug shall be a 3-blade male plug for use in receptacles described in ANSI C73.11. The lamp cord shall be inserted into the housing through a cutout in the frame, as indicated in DPSC Drawing No. 10173. A grommet shall provide strain relief at the cutout, so that a mechanical strain on the exposed lamp cord will not be transmitted to the wiring within the housing. The overall exposed length of the cord and plug shall be not less than 63 inches. The rating and current capacity of the cord and plug shall be not less than that of the lamp (see 3.2.4). The lamp cord shall be attached to the lamp assembly as indicated by DPSC Drawing No. 10173.

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3.2.6 "Off-on" switch. A 2-position "off-on" switch conforming to DPSC Drawing No. 10173 shall be supplied. The "off-on" switch shall be attached to a cutout in the frame of the housing, as indicated in the drawing. The "off-on" switch shall control the operation of the lamp (see 3.2.4).

3.3 Performance.

3.3.1 Lamp control. When tested as specified in 4.4.1, the "off-on" switch of the view box shall control the operation of the lamp, and the lamp shall operate without any indication of failure. The lamp bulb shall operate only when the "off-on" switch is in the "on" position.

3.3.2 Observation glass surface temperature. When tested as specified in 4.4.2, the observation glass on the view box shall reach and maintain a surface temperature within a range of 45° to 50°C.

3.4 Finish. The wood frame and housing shall be smoothly finished and shall also be suitably coated to resist moisture penetration.

3.5 Workmanship. Workmanship shall be first class throughout. The view box shall be free from defects which detract from its appearance or may impair its serviceability. The view box shall show no evidence of warpage. The wood frame and housing shall be free of burrs, chips, cracks or splinters. The housing shall be able to tilt freely in the frame. The observation glass shall be free of cracks, chips, blisters or scratches. When placed on a flat, level surface, the base of the view box shall show no evidence of rocking or wobbling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or

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purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. The unit of product shall be the individual view box.

Table I Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 or Table II, as applicable. The unit of product shall be the individual view box. The acceptance number shall be zero.

Table II Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level
Illumination	3.3.1	4.4.1	100%
Operating temperature	3.3.2	4.4.2	S-1

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4.3 Examination. Examination shall be conducted in accordance with the classification of defects listed in Table III. Examination shall not be restricted to the possible defects listed in the table.

Table III Classification of defects

Categories	Defects
Major	
101	View box not free of missing components.
102	Housing of view box not able to tilt freely without evidence of obstruction(s).
103	View box not free of warpage.
104	Stand or housing not free of burrs, cracks, chips or splinters.
105	Base of view box not free of rocking or wobbling when base is placed on a flat, level surface.
106	Components of view box not adequately secured to unit.
107	Metal components of view box not free of corrosion.
108	Observation glass not essentially smooth and flat.
109	Observation glass not free of cracks, chips, blisters, scratches and imbedded or unremovable particles.
110	Retaining strip not capable of being press-fitted into or removed from housing.
111	Moisture-protective coating not properly applied to wood surfaces.
112	Lumiline lamp not capable of being readily inserted in or removed from lampholders.
Minor	
201	View box not free of removable dirt, grease or foreign particles.
202	Identification marking not present, complete, legible or permanent.

4.3.1 Dimensional examination. The view box shall be examined for defects in dimensions. Any dimension not within specified limits shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Lamp control. A Lumiline lamp shall be positioned in the lampholders in the view box. The lamp cord shall be connected to a suitable power source. The two positions of the switch shall be checked for proper operation of the lamp bulb.

4.4.2 Observation glass surface temperature. The test shall be conducted with the lamp bulb and observation glass components of the view box at a room temperature of 25°C. ± 5°. The lamp bulb and observation glass shall be positioned in the view box housing. The lamp cord shall be connected to a suitable power source, and the lamp bulb shall be illuminated. After a 20-minute warm-up period, a suitable

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temperature-measuring device shall be used to determine the temperature on the observation glass (i.e. top side) at the following positions over the length of the lamp bulb:

a. The center of the observation glass
b. Three inches from both edges of the observation glass.

After the initial measurements, the temperatures shall be rechecked two more times at 30-minute intervals.

4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.5).

5.1.1 Level A. Each unit shall have the cover glass wrapped in tissue paper and inserted within its respective position on the view box. In addition, a piece of fiberboard cut to the size of view area shall be placed on top of the glass cover and secured with tape in such a manner as to immobilize the glass cover to prevent movement during shipment and handling. The unit shall be further packaged in a snug-fitting fiberboard box constructed in accordance with PPP-B-636, type CF, class domestic. In addition, the box shall be designed with interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the item. Closure of the box shall conform to method I or II as specified in PPP-B-636.

5.1.2 Level C. Units shall be packaged in standard commercial containers of

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the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.5).

5.2.1 Level A. Twelve units packaged as specified in 5.1.1 shall be packed in an exterior container designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.2 Level B. Twelve units packaged as specified in 5.1.1 shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method I or II as specified in the appendix of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 The view box covered by this specification is listed in the Federal Supply Catalog as indicated:

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National Stock No.	Item Identification
6630-00-299-9834	VIEW BOX, RH BLOOD TYPING, 115 volt, AC-DC

6.2 The view box is intended to be suitable for use with microscope slides listed in the Federal Supply Catalog as indicated below:

National Stock No.	Item Identification
6640-00-494-3893	SLIDE, MICROSCOPE, Plain, 25 x 75 mm.

6.3 The Lumiline lamp bulb is listed in the Federal Supply Catalog as indicated below (see 3.2.4):

National Stock No.	Item Identification
6240-00-504-2526	LAMP, INCANDESCENT, 115 volt, 40 watt, Bulb T-8, Inside Coated

6.4 The slide method Rh blood typing procedure is intended to be in accordance with the American Association of Blood Banks Technical Methods and Procedures Manual.

6.5 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Applicable levels of packaging and packing (see 5.1 and 5.2).

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6630-0241
is

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-V-36590A (DSA-DM) View Box, Rh Blood Typing, 115 volt, AC-DC

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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